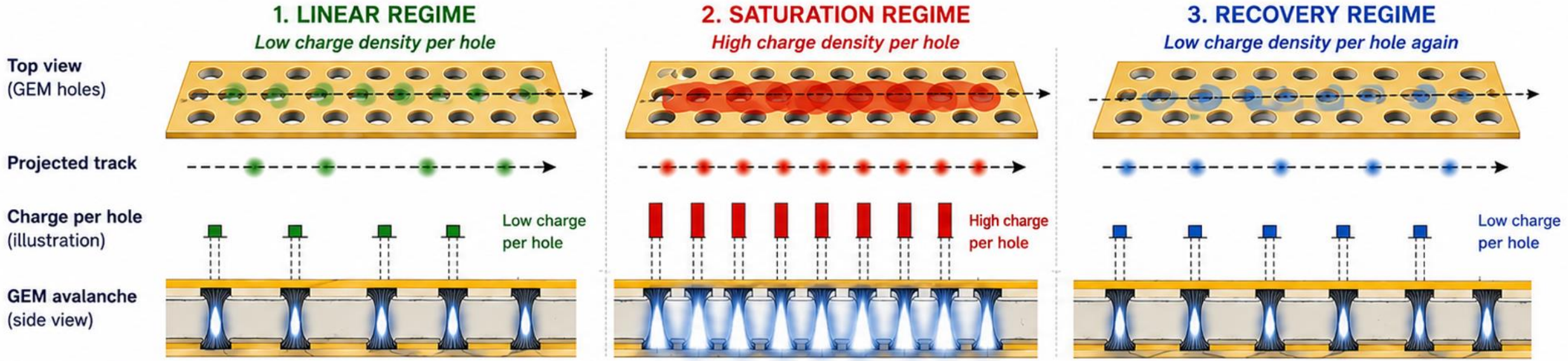
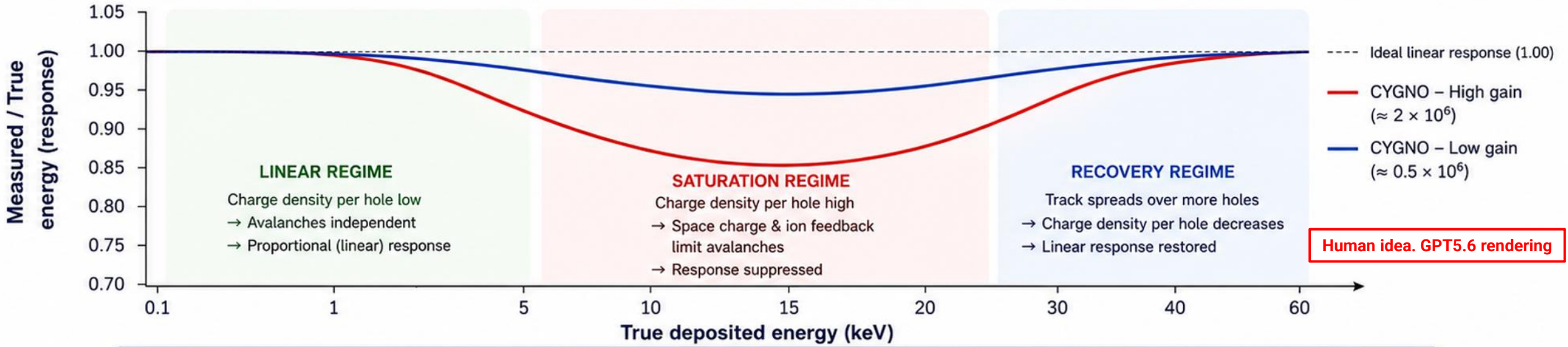
Saturation effect in CYGNO GEM-based detectors and correction at high gain

Davide Fiorina – GSSI & INFN LNGS

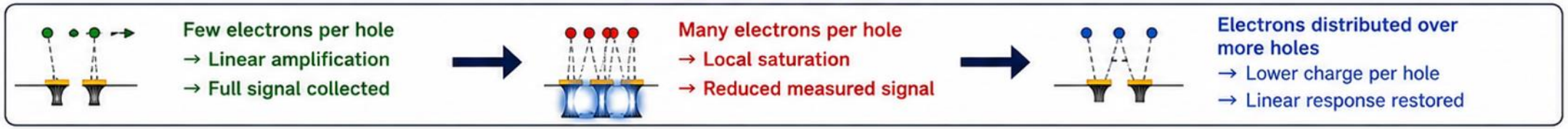
On behalf of the CYGNO collaboration

GEM RESPONSE CONTROLLED BY TRACK CHARGE (HOLE) DENSITY





Human idea. GPT5.6 rendering

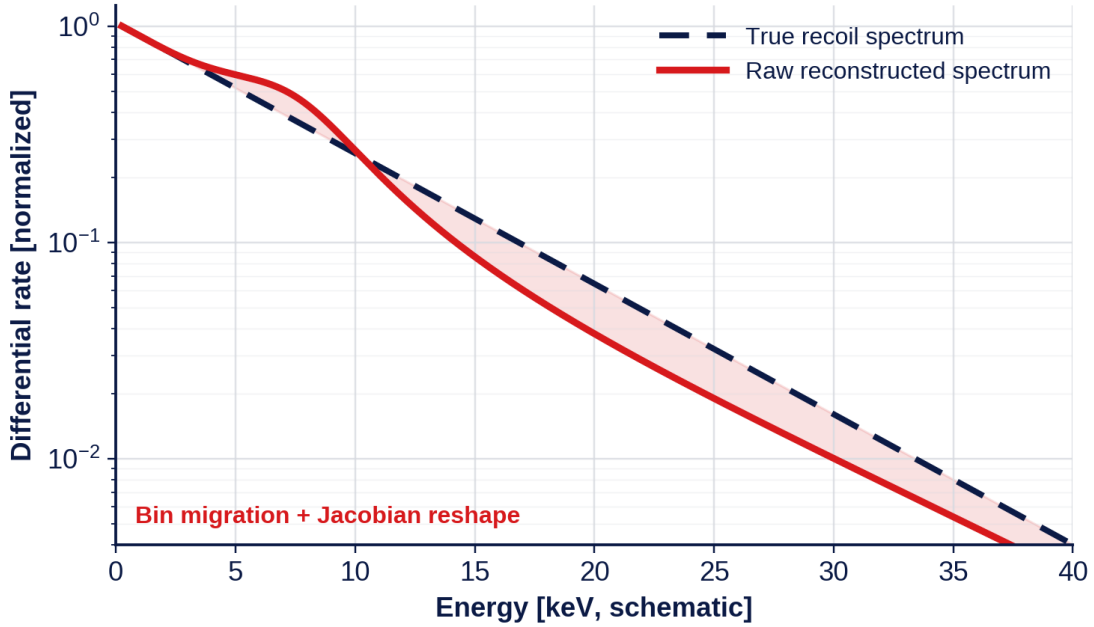


RAW SATURATION DISTORTS THE DARK-MATTER RECOIL SPECTRUM

$$dR/d_{Erec} = dR/dE_{true} \cdot |dE_{true}/d_{Erec}|$$

1. RAW RESPONSE

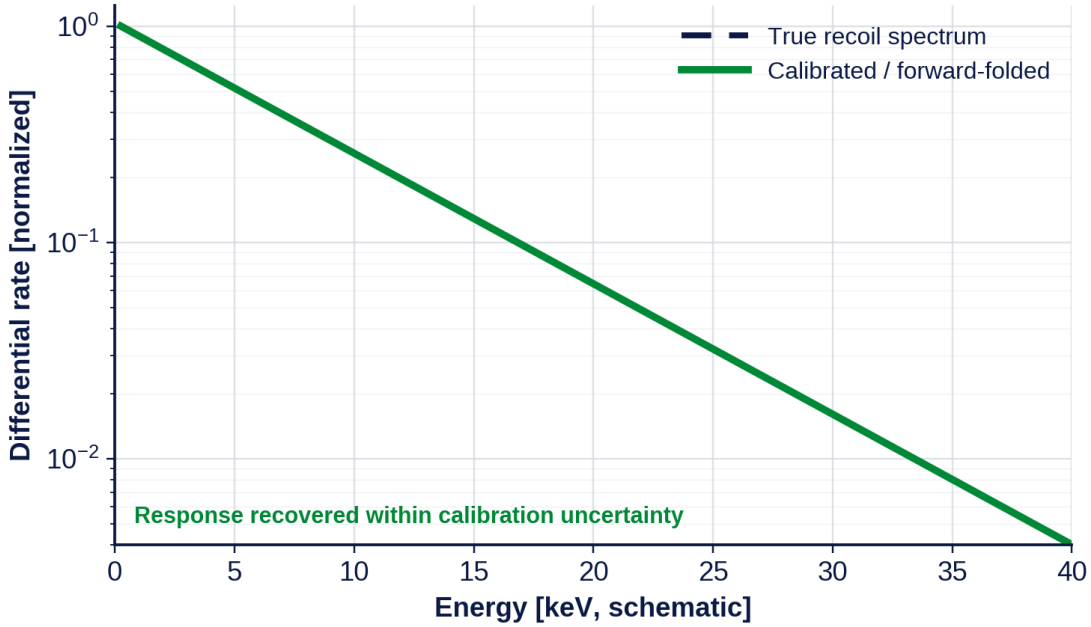
$E_{true} \rightarrow E_{rec} = f(E_{true})$



Shape / rate / inferred WIMP parameters biased

2. CALIBRATED RESPONSE

invert f or forward-fold the detector response



Spectrum recovered within calibration uncertainty

THE DELICATE POINT

Energy calibration works

if the same response function applies

→

$E_{rec} = f(E_{true}, Q_{hole}, G, topology)$

local response—not energy alone

→

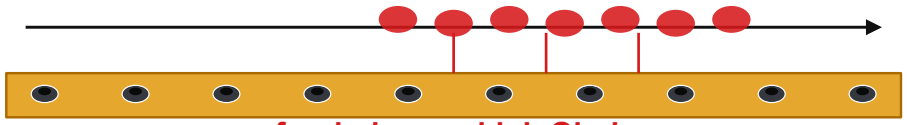
Do NR and ER saturate equally?

Human idea. GPT5.6 rendering

SATURATION CAN COLLAPSE NR / ER TOPOLOGICAL SEPARATION

NUCLEAR RECOIL

short, dense track

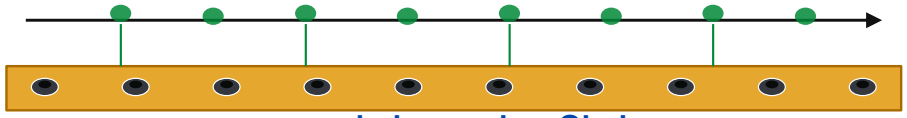


few holes → high Qhole

$Q_{hole}(NR) > Q_{hole}(ER)$

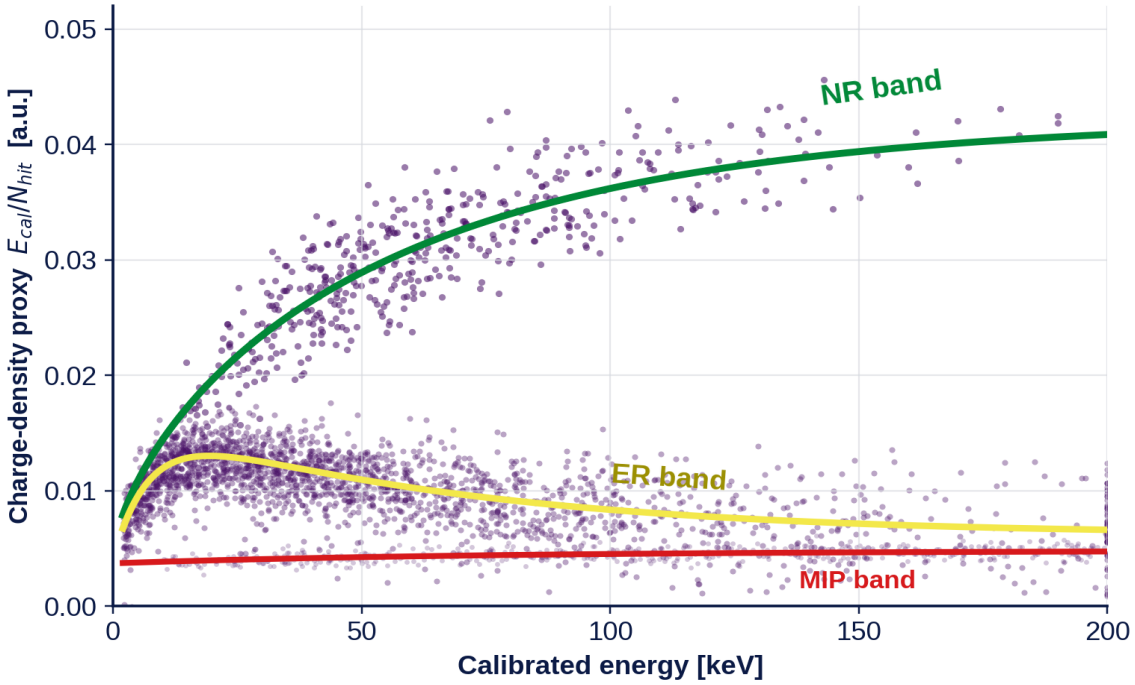
ELECTRON RECOIL

long, diffuse track

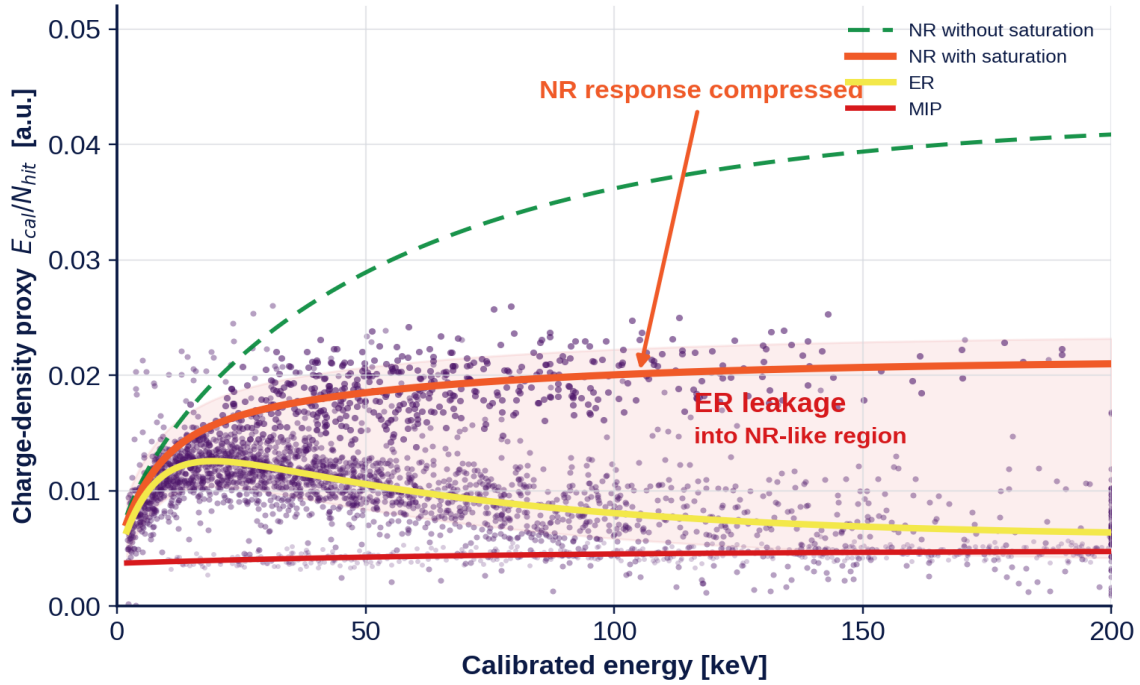


many holes → low Qhole

RESPONSE WITHOUT SATURATION



HIGH-GAIN SATURATION



Dense-track response ↓ → NR / ER band separation ↓ → ER leakage into the NR selection ↑

Human idea. GPT5.6 rendering

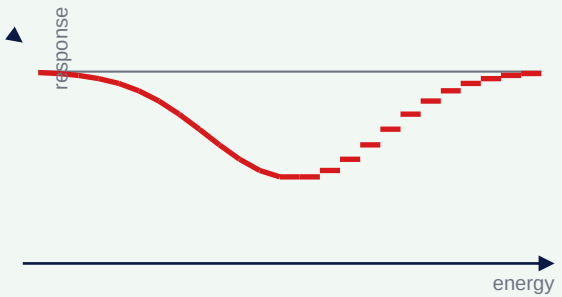
CYGNO STRATEGY: MODEL → CALIBRATE → MITIGATE

1. MODEL LOCAL RESPONSE

Qhole ≃ Qtrack / Nholes

track morphology | diffusion / z | gain G

$$R = R(E, Q_{hole}, G)$$

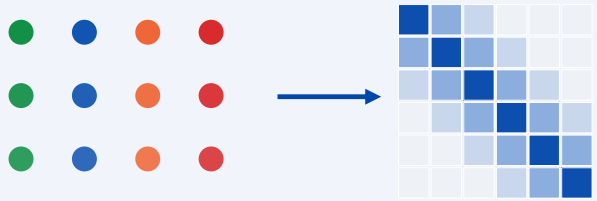


validated model

→ detector simulation

2. CALIBRATE THE PHYSICS SAMPLE

multi-E × z × topology



response matrix

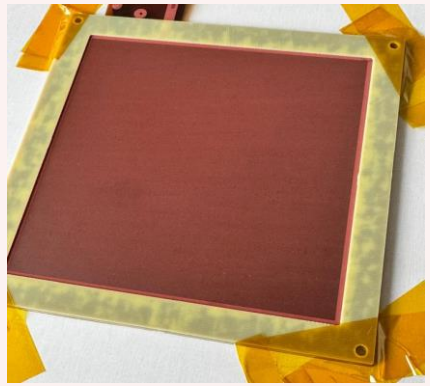
P(reco | true, topology)

correct E or forward-fold

DM likelihood

NR ↔ ER transfer must be measured

3. REDUCE SATURATION IN HARDWARE

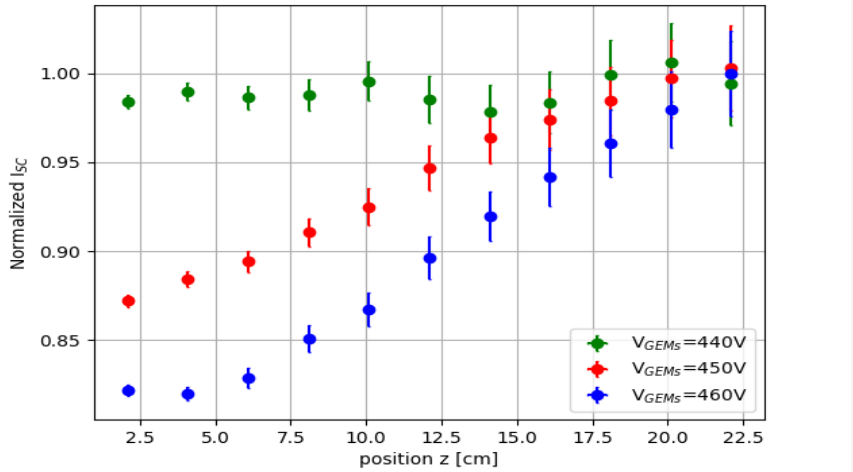


V-Bond GEM

>2× lower saturation

at comparable light output

Selected for CYGNO-04



normalized 55Fe response vs drift distance

Energy nonlinearity is correctable. The critical systematic is an unmodelled topology-dependent response.

Human idea. GPT5.6 rendering

V-Bond result and saturation model: supplied CYGNO saturation presentation; Amaro et al., EPJC (2026).